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Electric Vehicle Charging Station Recommendations Considering User Charging Preferences Based on Comment Data

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Abstract: User preferences are important for electric vehicle charging station (EVCS) recommendations, but they have not been deeply analyzed. Therefore, in this study, user charging preferences are identified and applied to EVCS recommendations using a hybrid model that integrates LightGBM and singular value decomposition (SVD). In the model, LightGBM is used to predict user ratings according to users' comments regarding charging orders, and the feature importance reported by each user is output. Then, a co-occurrence matrix between users and charging stations (EVCSs) is constructed and decomposed using SVD. Based on the decomposed results, the final evaluated scores of each user for EVCSs can be calculated. Upon ranking the EVCSs according to the scores, the EVCS recommendation results are obtained, taking into account the users' charging preferences. The sample data consist of 28,306 orders from 508 users at 241 charging stations in Linyi, Shandong, China. The experimental results show that the proposed hybrid model outperforms the benchmark models in terms of precision, recall, and F1 score, and its F1 score can be increased by 96% compared with that of the traditional item-based collaborative filtering method with charging counts for EVCS recommendations.

Keywords: electric vehicle charging station recommendation; user charging preference; user comment data; LightGBM; singular value decomposition

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1. Introduction

In recent years, electric vehicles (EVs) have gained popularity for their eco-friendly and energy-efficient characteristics. Advances in fast charging and lithium-ion batteries have propelled the transportation sector toward an electrified era led by EVs [1]. As a result, numerous countries are committed to the development and popularization of EVs [2]. China, as a pivotal participant in this endeavor, has positioned the advancement of EVs as a viable strategy to achieve the goal of carbon neutrality in the transportation sector. By June 2024, China had approximately 24.72 million new energy vehicles, accounting for 7.18% of the total vehicle population. The total electricity consumption for EVs has reached 242.9 billion kWh, leading to a year-on-year increase of 54.6%. The amount of charging piles in China has reached 10.244 million, including 3.122 million public and 7.122 million private piles, ensuring that the charging demand for EVs can be met. However, there is still considerable room for improvement in the popularity of EVs and the experience of charging services for EV owners [3].

The imbalance between the construction of EVCSs and the number of EVs is one of the key factors that restricts the increase in the popularity of EVs [4]. In order to improve the charging experience of EV users, personalized EVCS recommendations are of great significance [5]. An effective personalized EVCS recommendation considers uncertainty due to user preferences such as waiting time, charging costs, and driving distance, helping

users to make independent charging decisions. Meanwhile, the recommendation can also improve the utilization rate of EVCSs, optimize the allocation of charging facilities, and promote the sustainable development of the EV industry.

User comments are subjective data generated by EV charging users based on their own experiences, which can authentically reflect user preferences, provide insights into user needs, and enable personalized recommendations. However, in existing research on EVCS recommendations, there is a lack of in-depth analysis of user comment data. Therefore, in this study, we identify user charging preferences by mining charging order comment data from EV users and consider these learned user preferences in the EVCS recommendation ranking to provide more personalized results. First, in this study, user ratings of EVCSs are predicted based on historical charging orders and user comments, and user preference features are extracted using LightGBM. Then, a user–EVCS co-occurrence matrix is constructed, and singular value decomposition (SVD) is employed to predict the potential ratings of EVCS by users. As a result, EV users can receive EVCS recommendations that are highly aligned with their preferences, thus enhancing their experience with the EV charging service.

2. Literature Review

2.1. EVCS Recommendation

Research on EVCS recommendation focuses mainly on the analysis of the influencing factors, the construction of recommendation models, and the formulation of EVCS recommendation strategies [6–8]. These studies can be divided into two types. On the one hand, from the perspectives of the power distribution network, transportation systems, and charging infrastructure, EVCSs have mainly been recommended to reduce the risk of power grid overload and to alleviate traffic congestion issues [9–12]. On the other hand, from the user's perspective, previous research focused on recommending optimal EVCSs based on user needs such as range anxiety [13], charging costs [14], user preferences [15], and user privacy [5].

Existing studies have explored the EVCS recommendation problem using various data and methods. Research data include EV charging order data, EVCS data, EV data, and driving trajectory data. The methods include collaborative filtering, blockchain, and multi-objective optimization methods [5,11,15,16]. The data and methods are summarized in Table 1, which shows that EV charging orders were most frequently used in previous studies.

Table 1. Comparative study of EVCS recommendation research.

Year	Ref.	Data	Method	Problem
2019	[14]	Simulation data: 24 h EV data	Integer linear programming	EVCS recommendation based on distance, charging time, and charging cost
2019	[15]	Real data: track data from some vehicles in Hangzhou	Dynamic time warping method	EVCS recommendation considers user travel route preferences, EVCS locations, prices, and queueing information
2019	[17]	Real data: historical user charging data	Collaborative filtering	Customized EVCS recommendation and navigation based on collaborative filtering
2021	[18]	Real data: historical behavior data from EV users (geographic location, remaining electric quantity, charging type, etc.)	Collaborative filtering	EVCS recommendation based on collaborative filtering
2021	[19]	Real data: EVCS locations, EVCS IDs, pricing information, etc.	Blockchain technology and decision optimization	EVCS recommendation based on charging efficiency and safety

Table 1. *Cont.*

Year	Ref.	Data	Method	Problem
2022	[5]	Real data: EVCS data (historical data on the charging vehicle, cost, and latitude and longitude) and EV data (battery capacity, remaining battery, latitude and longitude, and charging time)	Vertical federated learning and blockchain technology	Recommendation of safe EVCSs based on cloudlet
2022	[11]	Simulation data: EV charging order	Graph reinforcement learning and collaborative filtering	Multi-objective fast EVCS recommendation based on the interests of power-transportation coupling system, fast EVCS, and EV users
2023	[4]	Real data: real-time availability record of EVCSs, charging prices, charging power, and charging request	Reinforcement learning	EVCS recommendation to minimize the overall charging waiting time, average charging prices, and charging failure rate
2023	[20]	Real data: historical user charging data	Restricted Boltzmann machine-learning algorithm and waterwheel plant algorithm	EVCS recommendation based on user preference
2024	[12]	Simulation data	Spotted hyena optimizer algorithm	EVCS recommendation based on stability and efficiency of distribution system infrastructure
2024	[13]	Simulation data: state of battery, arrival rate, charge and discharge rate, etc.	Analytic hierarchy process and multi-objective optimization	EVCS recommendation based on energy, total response time, charging cost, and battery degradation
2024	[16]	Real data: Foursquare check-in dataset	Tensor decomposition, mixed-integer linear programming and blockchain	Personalized EVCS and charging time recommendations for EVs based on blockchain
2024	[21]	Real data: type of facility, EV grade, estimated charging time, etc.	Analytical hierarchical process and technique for order of preference by similarity to ideal solution	EVCS recommendation based on user preference

Despite the achievements of the previous research, from the user's perspective, EVCS recommendations have not taken into account the user preference information hidden in the user comment data. Users' personalized charging needs have not been fully explored. Subjective user comments are valuable for understanding user needs, achieving personalized recommendations, and improving user experience. Therefore, in the EVCS recommendation system, the analysis and use of user comment data will accurately match users' charging needs with station characteristics, improving the accuracy of station recommendations and the quality of charging services.

2.2. User Preference Learning

User preference learning plays an essential role in recommendation systems. By analyzing users' historical behavioral data, comment data, rating data, and historical search data, it captures data on the individual's personalized needs and points of interest, thereby providing highly customized recommendation content [21,22]. Therefore, user preference learning has been used in recommendations for clothing [23], news [22], movies [24], and EVs [25], among others, and has achieved good personalized recommendation results.

The methods used to learn user preferences include questionnaires, rating matrix analysis, review text analysis, item and user interaction information analysis, and user group information analysis. For example, the authors in [26] present a method that is capable of providing clear preference information to improve collaborative filtering under

the conditions of sparse ratings or a small number of users. In [27], a new multifaceted user interest model that analyzes the comment text is proposed. To address cold start and data sparsity issues in item recommendations, the researchers in [28] use deep learning to extract user preferences embedded in review texts and rating data. In response to the problem that most existing studies use a single type of interaction to simulate the relationship between users and items, in [29], a multi-behavioral GNN model is proposed that extracts interaction information and user preferences from multiple types of user behaviors. In [30], the authors address the issue of check-in preference modeling in point-of-interest recommendations by proposing a recommendation method that constructs a unique hyper-spherical interest model for each user to describe their preferences. The researchers in [31] address the issue of aggregating different user preferences in group recommendations by modeling user influence and considering preference differences when users are alone or in groups.

Recommendation algorithms based on EV user charging preferences can provide personalized EVCS recommendations according to specific user needs and expectations, thus increasing user satisfaction and loyalty [21]. The methods used for identifying user charging preferences include questionnaires and user preference modeling based on historical charging behavior. In [25], the authors investigated the attitudes and preferences of potential users towards various technical, environmental, and policy aspects of EVs. Moreover, in [32], current battery EV users' preferences for different smart charging tariff designs were analyzed and BEV user groups with significant preferences were characterized. The authors in [33] conducted a preference survey and found that the travel time, travel cost, vehicle-related variables, and charging characteristics significantly influenced the selection and charging behavior of battery EV users. Surveys are an effective method used to understand the charging preferences of EV users. However, user preference modeling technology based on historical charging behavior is more able to capture users' charging habits through in-depth data mining and then provide personalized charging services. In [34], a framework to model charging behaviors is proposed that shows the risk attitudes and preferences of EV users based on the cumulative prospect theory. In [21], user preferences are integrated with the multi-attribute decision theory to propose a user-preference-based EVCS recommendation scheme.

Most studies identified preferences through questionnaires or by modeling from users' historical charging data. However, user comments, which contain data on user preferences, personal opinions, and other subjective value information, have been ignored. Therefore, in this study, we explore users' personalized charging preferences based on comment data and consider user preferences in EVCS recommendations to optimize the sorting of EVCSs based on a hybrid method with LightGBM and SVD. This approach can generate more personalized EVCS recommendation results and improve the user charging experience.

3. Methods

To recommend EVCSs based on users' charging preferences, we propose a method that combines data preprocessing, sample selection, user rating prediction, and recommendation list output (Figure 1). The details are described as follows:

1. Sample Construction: The collected raw data have multiple features, including user codes, charging start time, charging fee, EVCS IDs, and user ratings. Data preprocessing is performed to address the problems of missing values, outliers, and duplicate records. Given the subsequent prediction, users with a number of charging times more than N are selected as samples to train and test the prediction models. After selection, a set of features is constructed for the EVCS recommendation. The set comprises features from two perspectives, i.e., user charging order and charging station service. These features not only reflect user historical behavior, but also contain the multidimensional service characteristics of charging stations.
2. User Rating Prediction: On the basis of the screened samples, user ratings are predicted to analyze their charging preference. For this purpose, lingual user comments are first converted into numerical values, and then LightGBM is applied to forecast

these values for each user. By analyzing the feature importance (F_i) output using the LightGBM model, the varying charging preferences of users can be identified.

3. Charging Station Recommendation: After gathering information on the user's personalized preferences, SVD technology is used to output the EVCS recommendation results. Specifically, a user–EVCS co-occurrence matrix (U) is constructed based on the feature importance (F_i) of the user and historical order data. After determining a matrix dimension (N) for the decomposition of SVD and a length of the recommendation list (K), a personalized EVCS recommendation list is generated for each user.

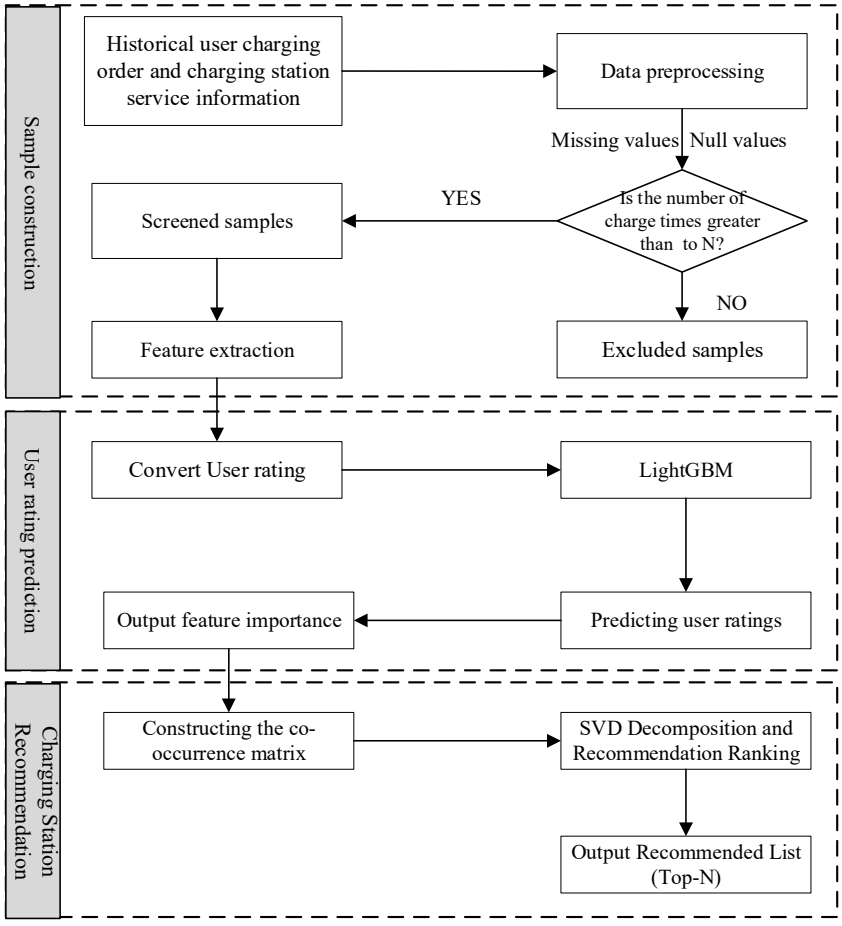


Figure 1. Method flow chart.

3.1. LightGBM

LightGBM is a gradient boosting algorithm based on the Gradient Boosting Decision Tree (GBDT) model [35]. It employs the construction of histograms to efficiently process the data and progressively improves the performance of the weak learner by fitting the model residuals. A leaf growth strategy is used to limit the depth of the tree. Iterative training is performed to split only the nodes with the largest gain, thus reducing the error and improving the prediction accuracy and speed.

Algorithm 1presents the pseudocode for the core algorithm of LightGBM, which outlines the process of building a decision tree using histograms. The algorithm takes the training data I, maximum tree depth d, and feature dimension mmm as inputs, and recursively splits the nodes layer by layer. For each node, a histogram is constructed for each feature by iterating over the data points. The histogram records the sum of the target values and the number of samples falling into each bin for the respective feature. Using this information, the algorithm evaluates the split gain for each feature and selects the feature with the highest gain and its corresponding split point to divide the current node. Feature

importance is updated by accumulating the gain associated with each split. After the split, the node set and data indices are updated, and the process continues until the maximum depth d or other stopping criteria are reached. The predicted value for each leaf node is computed as the mean of the target values within that node. The outputs of this pseudocode include the predicted scores for each data point and the feature importance scores.

Algorithm 1: LightGBM algorithm and pseudocode. Histogram-based algorithm

```
Input: I: training data, d: max depth
Input: m: feature dimension
nodeSet ← {0} ▷ tree nodes in current level
nodeSet ← {{0,1,2,...} ▷ data indices in tree nodes
for i = 1 to d do
  for node in nodeSet do
    usedRows ← rowSet[node]
    for k = 1 to m do
      H ← new Histogram()
      ▷ Build histogram
        for j in useRows do
            bin ← I.f[k][j].bin
            H[bin].y ←H[bin].y + I.y[j]
            H[bin].n ←H[bin].n + 1
        Find the best split on histogram H.
  Update rowSet and nodeSet according to the best
  gain, split ← EvaluateSplit(H)
  if gain > best_gain then
      best_gain, best_feature, best_split ← gain, k, split
  feature_importance [best_feature] += best_gain
   left, right ← SplitNode(usedRows, best_feature, best_split)
  nodeSet ← nodeSet ∪ {left, right}.
predictions← [0] * len(I)
for node in nodeSet do
  avg_prediction ← Mean (I.y[j] forj in rowSet[node])
  for j in rowSet[node] do
      predictions[j] = avg_prediction
Output: predictions, feature_importance
```

3.2. Collaborative Filtering Algorithm

Collaborative filtering is a recommendation algorithm based on association rules. It can be divided into user-based (UserCF) and item-based (ItemCF) collaborative filtering according to the objects of similarity. Given that the data used in this study contain more than N instances of user charging, there are sufficient data to analyze changes in user interests. Therefore, item-based collaborative filtering is employed. The specific steps are as follows.

Step 1: Based on historical charging order data, construct an $m \times n$ dimensional co-occurrence matrix U, where there are a total of m users and n charging stations, and u_{mn} represents the number of times that the user m went to the charging station n.

$$U = \begin{bmatrix} u_{11} & \dots & u_{1n} \\ \dots & \dots & \dots \\ u_{m1} & \dots & u_{mn} \end{bmatrix} \quad (1)$$

Step 2: Calculate the cosine similarity $cos(i,j) = \frac{i.j}{\|i\|\|j\|}$ between two of n column vectors of the co-occurrence matrix U, and construct the $n \times n$ dimensional item similarity matrix Γ.

Step 3: Based on the item similarity Γ, Top-k EVCSs are identified as the positive feedback in the target user's historical behavior to form a similar item set ψ.

Step 4: The items in the set ψ are sorted using similarity scores to generate the final recommendation list.

3.3. Singular Value Decomposition (SVD)

SVD has high generalization ability, low space complexity, and better scalability and flexibility compared to the typically collaborative filtering algorithm. The process of SVD is described as follows:

Step 1: Construct the co-occurrence matrix U based on the feature importance output by LightGBM. For a user, a charging order preference score is calculated by $P = \sum F_i * i$. For this user, the order preference scores for each charging station are added up to obtain the elements of the user–charging station co-occurrence matrix U.

Step 2: For a given co-occurrence matrix U, it can be decomposed into three submatrices:

$$U = R \sum V^T \tag{2}$$

where U is the user identity matrix with dimension $m \times k$. m is the number of users and k is the number of features. $\sum$ is the singular value diagonal matrix with dimension $k \times k$. V is the charging site identity matrix with dimension $n \times k$. n is the number of charging sites.

Step 3: Use the decomposed matrix to predict the final scores for EVCSs:

$$\hat{U}_{i,j} = R_i \sum V_j^T \tag{3}$$

where $\hat{U}_{i,j}$ is the predicted score of user i for charging site j; R_i is the row vector of user i in the feature matrix U; and ${V_j}^T$ is the column vector of charging site j in the feature matrix V.

Step 4: Sorting the SVD scores from high to low, Top-N EVCSs are selected as the recommendation results for users.

4. Experiments

4.1. Data

To predict user ratings, the rating information feature is used as the predicted variable. However, the original rating information is in text form and cannot be directly predicted. Therefore, textual data were converted into a numerical variable in this study. For this purpose, the words in the rating text were classified as positive and negative. There were 8 positive words and 20 negative words, as shown in Table 2. Positive words were assigned a value of 1 point, and negative words were assigned a value of −1 point. Based on these definitions, the user rating for each order was calculated. Additionally, if the rating information was empty, it was assigned a value of 0.

Table 2. Positive and negative words in the user comment text.

Positive Words	Negative Words	Negative Words	Negative Words
No queuing	Wrong price	Slow charging	Abnormal stop
Affordable price	High price	Occupy a large space	Slow QR code scanning
Free parking	Wrong location	Startup failure	Incomplete guidance
Fast charging	Poor charging experience	No navigation	Long queuing time
Enough charging piles	Faulty piles	Unable to charge	
Accurate navigation	General environment	Poor environment	
Fast QR code scanning	Equipment failure	Need to wait	
Good environment	Frequent connector error	Unable to pull out connector	

Considering that rating prediction requires a certain amount of data for learning, we selected users with at least one nonempty charging order comment and more than 30 charging transactions. After selection, we obtained 28,306 transaction records from 508 users at 241 charging stations. To predict user ratings, 17 features were constructed based on the user order information and station data, including 14 features and 3 multi-category features (Figure 2). The multi-category features were the charging method, order source, and reason for transaction ending.

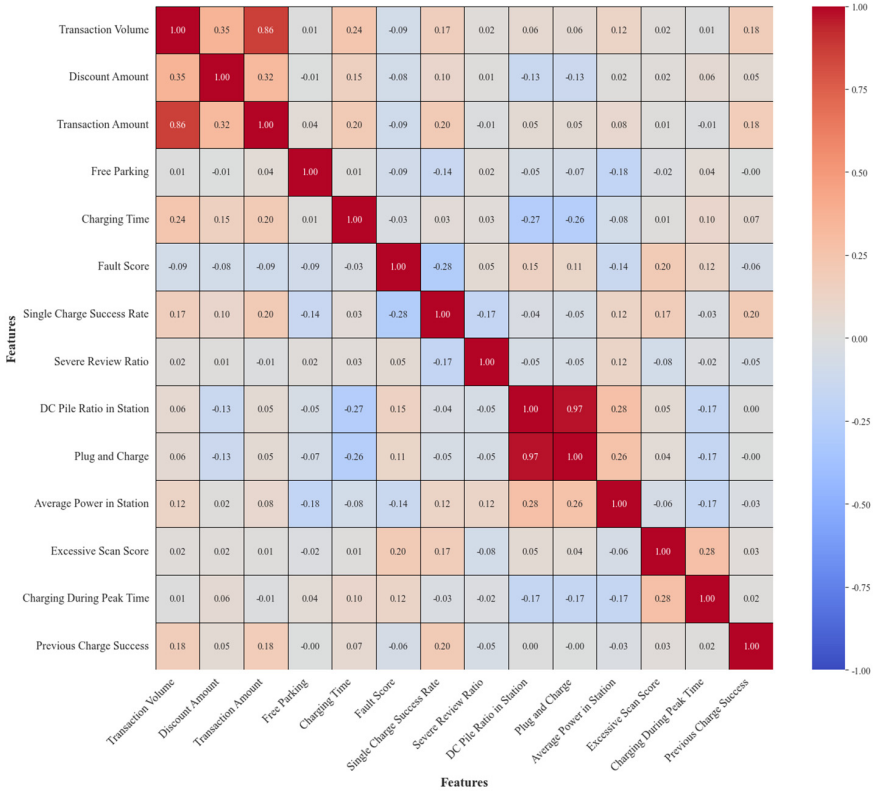


Figure 2. Heat map of 14 features for user charging rating prediction.

4.2. Experimental Setup

4.2.1. Experimental Setup for User Rating Prediction

To verify the performance of the user rating prediction, 80% of the processed dataset was used as the training set, and 20% was used as the test set (Figure 3). Since most user ratings in the dataset were 0, stratified sampling was employed when dividing the training and test sets. This ensured that different rating samples were included in both sets, facilitating diverse learning during model training.

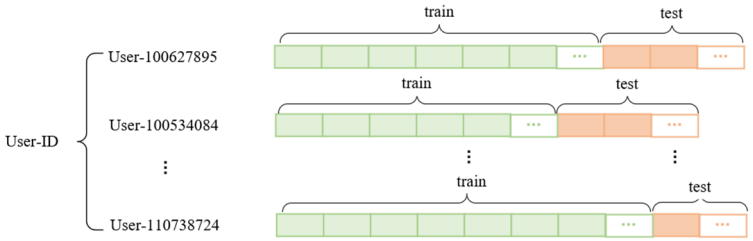


Figure 3. Training and testing set partition diagram.

In order to analyze the performance of different models in user rating prediction, three prediction models were used: LightGBM, XGBoost, and Random Forest. The parameter settings for each model are shown in Table 3.

Table 3. Main parameter settings for three prediction models.

	LightGBM	**XGBoost**	**Random Forest**
Learning_Rate	0.01	0.01	0.01
N_Estimators	200	200	200
Max_Depth	10	10	10
Random_State	42	42	42
Min_Samples_Split	-	-	2
Min_Samples_Leaf	-	-	1
Sample_Weight	(0,1,2)	(0,1,2)	(0,1,2)

4.2.2. Experimental Setup for Charging Station Recommendation

For charging station recommendations, we divided the dataset into training and test sets in chronological order. The first 80% of the data from users were used as the training set, and the remaining 20% were used as the test set. The training set was utilized for model training, while the test set was employed to evaluate the recommendation performance of the model.

To obtain station recommendation results, we decomposed the covariance matrix using the SVD algorithm and then calculated the cosine similarity between the charging stations based on the decomposition results. The final recommendations were the most similar stations. To determine the optimal decomposition length, the model's recommendation performance was analyzed using different matrix decomposition dimensions (K), ranging from 50 to 200. For the optimal length of the recommendation list, the model performance was analyzed using different lengths of the recommendation list (N), ranging from 1 to 10.

To verify the effectiveness of LightGBM in learning user charging preferences and the superiority of the SVD method, we adopted the ItemCF method in this study for the comparison of recommendation performance. First, the number of user charging times at a station was taken as an element of the user–charging station co-occurrence matrix to characterize user preferences. Then, the SVD method and the ItemCF method were used to extract user preferences for charging stations to obtain the recommendation results, and, finally, the recommendation performance was compared.

4.2.3. Performance Evaluation

To evaluate the model performance, we used four prediction criteria and three classification evaluation criteria, including the mean square error (MSE), mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), precision, recall, and F1 score. These criteria are defined below:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{4}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{5}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y_i})^2} \tag{7}$$

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

where y_i is the true value; $\hat{y}_i$ is the predicted value; n is the number of samples; TP is the number of true positives—i.e., charging stations that the model correctly predicts to be of interest; FP is the number of false positives—i.e., charging stations that the model predicts to be of interest but are not actually interested; and FN is the number of false negatives—i.e., charging stations that the model does not predict to be of interest but are actually of interest.

4.3. Experimental Results and Discussion

4.3.1. User Rating Prediction Results

In this study, three mainstream machine-learning models were utilized for user rating prediction: LightGBM, XGBoost, and Random Forest. The results are illustrated in Figures 4 and 5. Figure 4 displays the distribution of the prediction results for the three models, allowing for the visualization of prediction accuracy and bias. The optimal prediction lies on the diagonal, where the actual score is equal to the predicted score. The concentration of prediction values around this diagonal reflects the model's prediction accuracy. The overall results indicate that all three models deviate from the diagonal to some extent, with some user ratings exhibiting poor prediction performance. The reasons for these discrepancies are analyzed in Section 4.3.4. Among the models, LightGBM shows the highest concentration of prediction points around the diagonal, significantly outperforming Random Forest and XGBoost and demonstrating superior prediction accuracy. For further comparison, Figure 4 shows the MSE, MAE, RMSE, and R^2 of the models. LightGBM has the best prediction performance with the lowest MSE (0.9446) and MAE (0.2969), and the highest value (0.2935). Therefore, subsequent analyses are based on the user rating prediction model constructed with LightGBM.

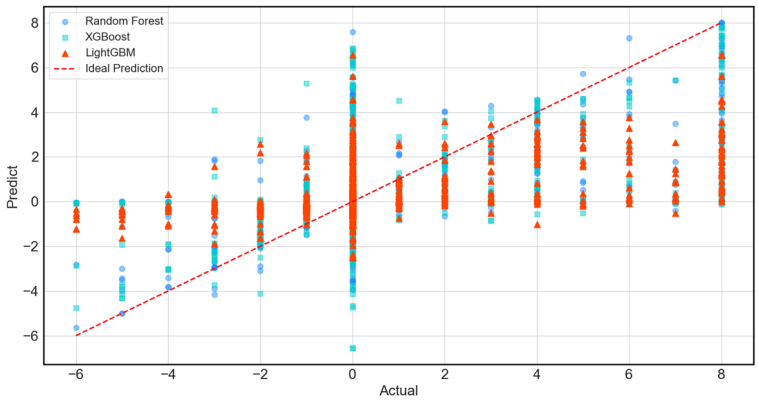


Figure 4. Results of user rating prediction of models.

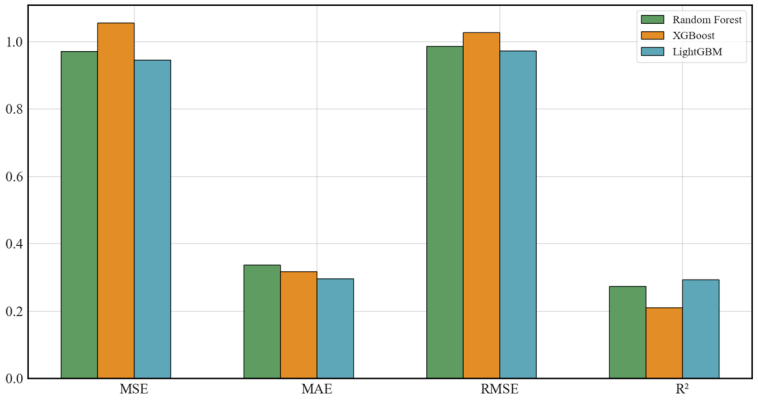


Figure 5. Performance of model user rating prediction.

4.3.2. User Preference Learning Results

Given the superior predictive performance, we then analyzed the user charging preferences based on the feature importance derived from the LightGBM model. The overall feature importance results are illustrated in Figure 6. Among all features, 'charging time', 'transaction amount', 'transaction power', 'offer amount', and 'reason for transaction end' have the greatest impact on predicting user ratings. These features play a key role in the user rating prediction model. On the contrary, features such as 'free parking', 'charging method', 'excessive code scanning times', and 'whether charging occurs during peak or off-peak hours' have less impact on user ratings. Although these features may influence user behavior to some extent, their limited contribution to rating prediction may be due to users' lower sensitivity to these factors or their minimal variability in ratings.

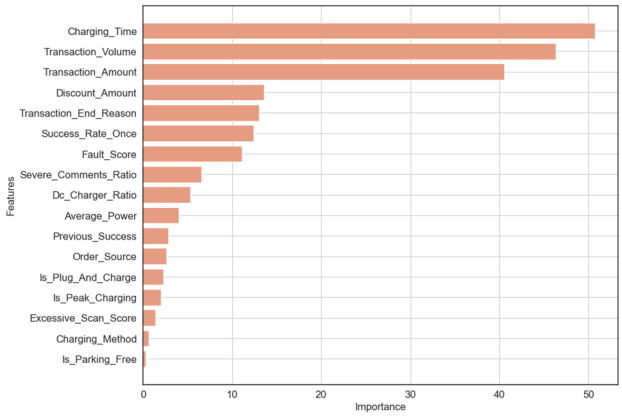


Figure 6. Overall feature importance from LightGBM.

The importance varies among different users. Specifically, all feature importance is zero for 304 users, indicating that their charging preferences cannot be inferred from the charging rating prediction. The possible reasons for this are discussed in Section 4.3.4. In contrast, personalized charging preferences were identified for 204 users through a feature importance analysis. For instance, for User 100266869, as illustrated in Figure 7, the 'fault score' feature was the most significant. User 100266869 was charged 78 times over the past six months. Moreover, 16 transactions were terminated due to failures at the charging station or post, making the 'fault score' a critical factor in predicting this user's ratings. For User 107935466, the 'charge time' feature was most significant due to the average charging time of 81 min. The average charging time was significantly higher than the overall average of 42 min, thereby making this feature crucial for predicting User 107935466's charging preferences.

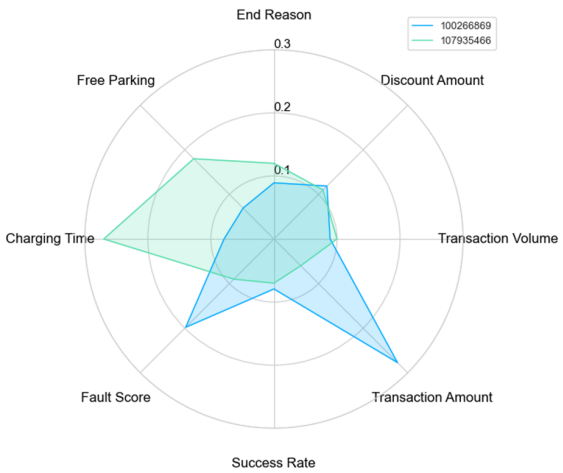


Figure 7. Example of users' personalized charging preferences.

4.3.3. Results of the Charging Station Recommendations

In EVCS recommendation, the dimensions of matrix decomposition (K) and the length of the recommendation list (N) significantly affect the results. To determine the optimal K and N, we conducted experiments to compare the model performance across different values of K and N. The results are shown in Figures 8 and 9. The experimental findings in Figure 8 indicate that the accuracy, recall, and F1 score of the model significantly improve when K ranges between 50 and 90. In particular, the F1 scores stabilize once K exceeds 90, with the optimal model performance achieved at K = 160. In Figure 9, the recall substantially increases when N is between 1 and 4. Although the model performance continues to improve with N greater than 5, the enhancement is relatively small. Therefore, to consider the computational efficiency and practical applications, we selected a recommendation list length of 5 as the final model setting.

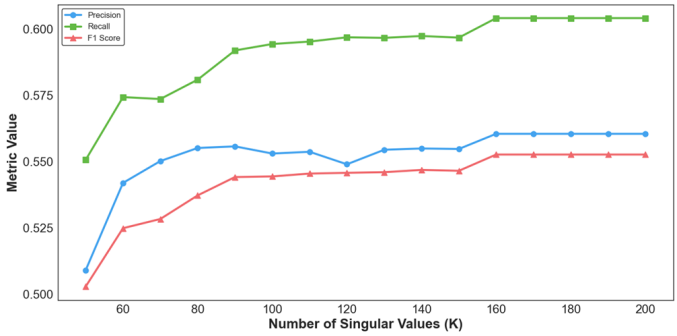


Figure 8. Model performance with different decomposition dimensions.

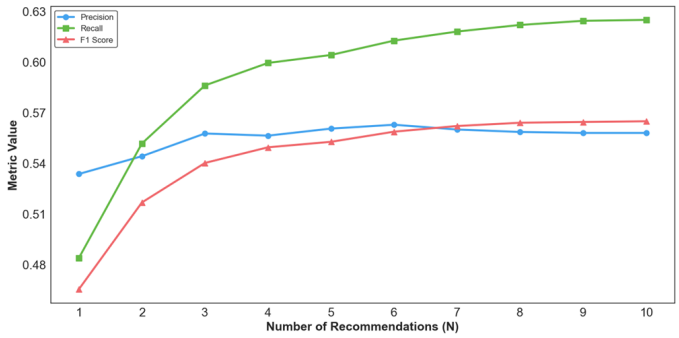


Figure 9. Model performance with different recommendation list lengths.

In the case of N = 5, Table 4 presents examples of the final list of recommendations. For user 1, the recommended EVCSs are numbered 21, 132, 78, 182, and 120, and the scores of the recommended EVCSs are 3.66, 3.57, 1.91, 1.24, and 0.85, respectively.

Table 4. User charging station recommendation results.

User	Charge Station and Predicted Score
User 1	(21, 3.66), (132, 3.57), (78, 1.91), (182, 1.24), (120, 0.85)
User 2	(12, 4.19), (200, 4.07), (198, 1.88), (78, 1.86), (15, 1.43)
User 3	(106, 14.28), (231, 3.69), (175, 1.42), (123, 1.07), (24, 0.32)

For the EVCS recommendation, we constructed and evaluated four models, and the results of the recommendation performance are shown in Figure 10. The results demonstrate that SVD combined with LightGBM outperforms the other models, achieving a precision of 0.5604, a recall of 0.6040, and an F1 score of 0.5527. These performance criteria are higher than those of the other models, highlighting the superiority of the

proposed method in aligning users' charging preferences with the characteristics of charging stations. The SVD + LightGBM and SVD + Charge Count models exhibit a higher recall, accuracy, and F1 scores compared to the ItemCF + LightGBM and ItemCF + Charge Count models, respectively. These findings indicate the effectiveness of the SVD method in capturing the underlying relationship between user charging preferences and charging station features compared to ItemCF. Furthermore, the recall, accuracy, and F1 scores of SVD + LightGBM and ItemCF + LightGBM surpass those of SVD + ChargeCount and ItemCF + ChargeCount, respectively. These indicate the advantage of LightGBM in recognizing user charging preferences.

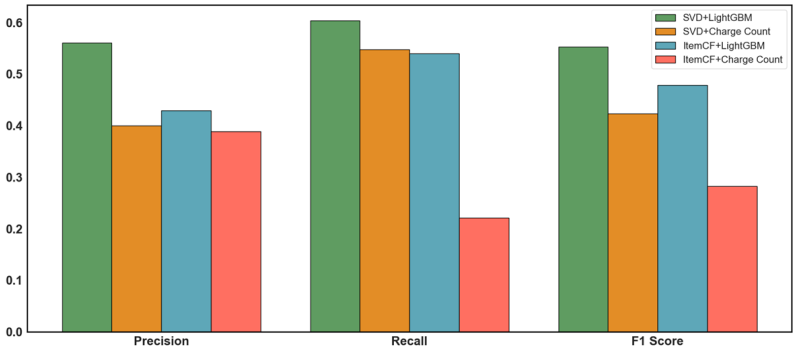


Figure 10. The accuracy, recall, and F1 score of different recommendation models.

4.3.4. Discussion

Based on the data characteristics of the selected sample, this section presents an analysis of the poor rating prediction performance and the zero importance of all features of some users. Specifically, poor predictive performance is related to a high imbalance in the data. The dataset contains 28,306 orders, of which 25,308 have zero ratings, indicating a severe imbalance. This imbalance causes the model to encounter predominantly negative samples (i.e., ratings of 0) during training, leading to the neglect of features and patterns associated with positive samples (i.e., non-zero ratings). Consequently, the model may default to predicting zero for all ratings to minimize the prediction error. Although this strategy might show favorable performance on training data, it can significantly impair evaluation metrics such as recall and F1 scores in practice. For example, User A has 50 orders in the dataset, 48 of which have a rating of 0, while only 2 orders have ratings of 2 and 5. As a result, the rating prediction model of User A achieves an accuracy of 0.96 but exhibits a recall of only 0.05 and an F1 score of 0.09.

There is also a significant imbalance in the evaluation data for 304 users with zero LightGBM outputted importance of all features. Among these users, 95.24% of their order ratings are zero. This high level of imbalance can obscure the actual contribution of features because the model primarily learns from non-zero samples during training. As a result, zero samples may not adequately reflect their impact on the model, leading to an underestimation of some feature importance.

Since the majority of user ratings in the dataset are zero, we applied oversampling to order the data from each user. Additionally, we adjusted the algorithm's weights to place greater emphasis on non-zero samples. As shown in Figure 11, the predictive performance of the LightGBM model improved slightly after implementing data imbalance handling techniques. Specifically, the MSE, MAE, and RMSE exhibited slight reductions, while the R^2 value increased significantly. Furthermore, the F1 score of the subsequent recommendation results also improved. This indicates that the data-imbalance-handling techniques used here can better capture user preferences for EVCS, thereby improving the recommendation outcomes.

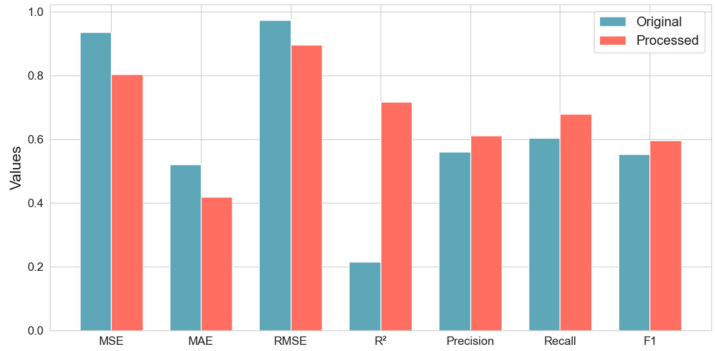


Figure 11. Changes in rating prediction results (MSE, MAE, RMSE, and R^2) and recommendation results (precision, recall, and F1) after data imbalance processing.

Additionally, user behavior preferences may change over time. Therefore, we analyzed the changes in the existing dataset. Specifically, we input the first 30 charging orders of each user into the predictive model as a baseline and output feature importance results after adding each subsequent order for incremental learning. As shown in Figure 12a, the overall standard deviation of feature importance is quite small, with most deviations concentrated within the range of [0, 0.009]. For the user with the largest changes in feature importance, as illustrated in Figure 12b, there are minor fluctuations in feature importance with the addition of order data; however, the overall trend remains stable.

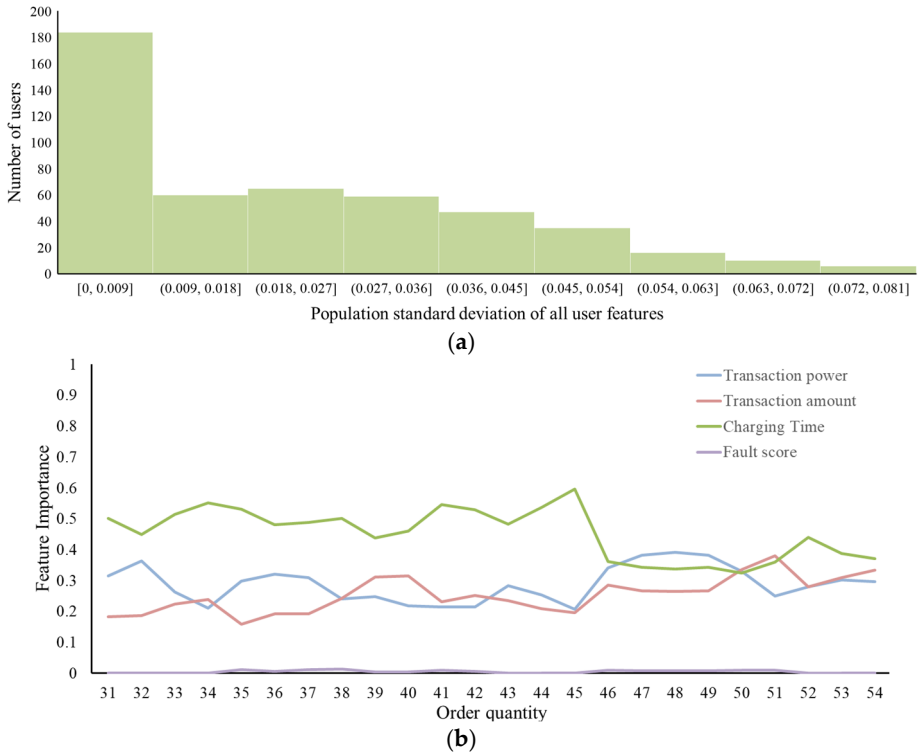


Figure 12. (**a**) Overall standard deviation of feature importance in incremental learning; (**b**) illustrations of feature importance changes for user with maximum feature importance variance.

According to the changeable charging preference, as in the examples shown in Figure 12b, we implemented dynamic EVCS recommendation experiments. The results showed that the recommended Top 5 EVCS remained the same for each user. The recommendation results do not change when applying the changeable charging preference. This indicates that, although the used data have dynamic changes, the time span is not

long enough to induce a significant impact of the charging preference learning on the final recommendation results. A longer time span will be considered in future studies.

As our model expands, particularly with large-scale datasets or real-time applications, computational complexity becomes a challenge due to the resource-intensive nature of feature extraction, training, and prediction. Another limitation that needs addressing is the computational complexity of the model. While LightGBM is effective for large-scale data, it still faces challenges in terms of training time and memory consumption when applied to highly imbalanced datasets. Future research could explore optimization techniques such as distributed computing, model pruning, or other scalable methods to reduce computational overheads while maintaining predictive performance.

5. Conclusions

In this study, a hybrid model was developed that integrates LightGBM and SVD to address the problem of learning user charging preferences and intelligent EVCS recommendations, using historical user charging orders and comment data. The experimental results demonstrate the effectiveness of the LightGBM model in learning user charging preferences. It can extract complex information from historical order data, significantly enhancing prediction accuracy and providing differentiated user charging preference data. For EVCS recommendations, we employed the SVD method in this study to decompose the co-occurrence matrix between users and charging stations. The results reveal that SVD effectively captures the latent relationships between them. By combining SVD with LightGBM, the proposed method outperforms other models in terms of precision, recall, and F1 metrics. Compared with other methods, the F1 score of the hybrid model is increased by 15%~96%. The accuracy of personalized EVCS recommendations is significantly enhanced. Further, we used oversampling and algorithm weight adjustment methods to handle the data imbalance problem and incremental learning for the problem of changeable charging preferences. The results show that data imbalance processing can enhance the EVCS recommendation results, while the learned changeable charging preferences do not have a significant influence on EVCS recommendations. This can be explained by the relatively short data collection time span.

This study describes an effective approach for personalized EVCS recommendations based on the learning of users' charging preferences. However, there remain challenges, including the need for more accurate prediction results and effective dynamic charging preference learning, as well as high computational complexity. Future research should expand the feature set to address the various factors influencing user ratings. Moreover, utilizing a dataset over a longer time frame and incorporating the dynamic aspects of user behavior through a time series analysis or regular update mechanisms could enhance the model's adaptability to changes in user preferences, further improving user preference learning and the performance of EVCS recommendations.

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